مدارس الإمارات الوطنية
Emirates National Schools

Design Department Weekly Lesson Plan

Dates: Oct 26 – 30, 2026 (Term 1 · Week 9) Grade/Class: MYP Year 2 (Grade 7) Teacher: Tichaona Gozo

Unit Title	Lesson Title	
Designing Ethical AI Systems – Developing Ideas (Criterion B)	**Lesson 1 (W9.1): Generating Two AI Design Ideas** **Lesson 2 (W9.2): Creating the AI Blueprint / Storyboard (Draft)**	
Learning Targets	**Success Criteria**	**Assessment tools and strategies** Formative / Summative
Lesson 1 (W9.1) – Students will be able to: • Sketch two distinct AI design ideas for their school problem, each with at least 4 annotations (input, process, output, user). • Compare both ideas against at least 3 design specifications from W8.2 using a comparison table. • Score each idea's feasibility (1–3) on time, resources, accuracy and fairness, and write one trade-off. **Lesson 2 (W9.2) – Students will be able to:** • Choose one idea and justify the choice with reference to at least 2 design specifications. • Draft a blueprint or storyboard showing at least 5 steps in order, from input to output. • Label at least 3 data inputs and 2 outputs, and mark 1 fairness or safety check.	**Lesson 1** • **MUST:** I can produce two distinct AI design ideas. • **SHOULD:** I can compare the ideas against my design specifications. • **COULD:** I can identify a trade-off between the two ideas. **Lesson 2** • **MUST:** I can show the basic steps/flow of my AI system. • **SHOULD:** I can label key data inputs and outputs. • **COULD:** I can show where a fairness or safety check occurs in the process.	**Lesson 1 – Formative 2 of 2 (Criterion B)** Two AI Design Ideas worksheet – teacher feedback against Criterion B strand ii (unweighted; feeds the Blueprint summative). **Lesson 2 – Ongoing formative** Teacher observation, questioning, peer feedback (two stars and a wish) and blueprint draft check (unweighted). Exit tickets (3-2-1) and confidence check in both lessons.
Standards	**ATL Skills**	**National Identity Links**
MYP Criterion B (Developing Ideas): strand ii – a range of feasible design ideas (L1); strands iii–iv – chosen design justified with the specification and planning drawings (L2). **CS.MS.5.2** The behaviour of AI systems reflects both the goals of the designers and the data used to train the system.	• **Thinking – creative:** generating distinct AI ideas • **Thinking – critical:** feasibility scoring, trade-offs, bias checks • **Communication:** annotated sketches and blueprints others can interpret • **Collaboration:** think–pair–share and	Students design an ethical AI system that solves a real problem in a UAE school (e.g. Lost & Found Helper, Library Book Recommender), showing how it protects privacy, avoids bias and benefits the school community. Technology, innovation, artificial intelligence and ethical digital design support the UAE's vision for a secure, innovative, knowledge-based economy; responsible

CS.MS.1.1 Complex problems can be broken into smaller parts to facilitate program implementation and revision. **Statement of Inquiry:** *Understanding how AI systems function through data and algorithms allows designers to evaluate their accuracy, fairness and ethical impact on users and communities.* **Key concept:** Systems · **Related:** Function, Evaluation · **Global context:** Scientific & technical innovation	peer feedback **GRASPS link:** students act as AI Designers creating the AI System Blueprint (input → process → output) for the Criterion B summative.	digital citizenship, critical thinking and future-ready skills. **Cross-curricular:** Computer Science

Technology being used – physical, websites, Learning Support Platforms, Apps, etc.

Own devices, Toddle, Mentimeter, School AI, Lesson 1 & 2 PowerPoints, Lesson 1 & 2 student worksheets, AI Disaster Cards (recap), coloured pencils.

Main Teaching and Learning Activities – Lesson 1 (W9.1): Generating Two AI Design Ideas

Bellringer	**Spec Check-In: Which Risk Does It Stop?** *(Think–Pair–Share)* Students reread the 3 design specifications they wrote in W8.2. In pairs, they decide which AI Disaster Card risk each specification protects users from: • Biased decisions • Data leak • Users don't trust it • Wrong decisions Sentence frame: "My specification ____ protects users from ____ because ____." Students absent from W8.2 use the example specifications on the worksheet. This links strand i (specifications) to today's strand ii (design ideas).
Time	5 min
Main Activities	**Mini-lesson 1 – What makes two ideas distinct?** (7 mins) The teacher introduces four AI forms: Software/App AI, Camera-based AI, Voice/Chatbot AI and Robot + AI. Ideas are distinct when they use a different input, process or way users interact (e.g. app recommender vs camera kiosk), not the same app in a different colour. **Worked example** (4 mins) Library Book Recommender: Idea A "BookBuddy" app vs Idea B "ShelfScan" camera kiosk, each annotated with Input, Process, Output and User. **Mini-lesson 2 – Is it feasible? Four lenses** (4 mins) Students learn to score each idea 1–3 for Time, Resources, Accuracy and Fairness & Privacy (max 12). The teacher models scoring BookBuddy (10/12) and ShelfScan (7/12). **Task 1 – Generate two AI design ideas** (15 mins) – Worksheet Part B Students choose, imagine, sketch and annotate Idea A and Idea B, adding at least 4 annotations each (input, process, output, user). The

	teacher circulates, asking: "How is Idea B really different? Where does the data come from? Would the audience understand your sketch?" **Tasks 2 & 3 – Compare and score feasibility** (10 mins) – Worksheet Parts C–D Students rate each idea against their 3 specifications (□ fully / ◐ partly / □ not met), complete the feasibility scorecard (/12) and write a conclusion sentence. **Task 4 – Spot the trade-off** (5 mins) – Worksheet Part E (COULD) Students use the frame: "Idea ___ is better for ___, but Idea ___ is better for ___. If I choose Idea ___, I will reduce the downside by ___." Challenge-level students predict how each idea could produce a biased result (CS.MS.5.2).
Time	45 min
Plenary	**Exit Ticket 3-2-1 + confidence check** • 3 features that make your two ideas different • 2 design specifications your stronger idea meets • 1 trade-off, or 1 question you still have Students rate themselves 1–3 against MUST/SHOULD/COULD. The teacher uses the ratings to group students for Lesson 2.
Time	5 min
Resources	Lesson 1 PowerPoint (W9.1), Lesson 1 student worksheet, W8.2 design specifications, picture cards / GRASPS example cards (Support), coloured pencils, devices, Toddle.
Main Teaching and Learning Activities – Lesson 2 (W9.2): Creating the AI Blueprint / Storyboard (Draft)	
Bellringer	**Fix the Flow: Lost & Found Helper** *(pairs)* Six jumbled step cards (A–F) for a Lost & Found Helper AI. Students order them from START to END and identify the step that keeps students' data safe. **Answer:** B → D → E → C → F → A. Step D (blur faces and names) is the privacy check and happens before the AI processes the photo.
Time	5 min
Main Activities	**Task 1 – Choose your idea and justify it** (7 mins) – Worksheet Part B Using their Lesson 1 comparison table and feasibility scores, students choose ONE idea and justify it against at least 2 design specifications (Criterion B strand iii). The teacher shows a model answer. **Mini-lesson – How AI works: Input → Process → Output** (5 mins) Inputs (photos, text, voice, sensor data), process (algorithm/model finds patterns and decides) and outputs (message, recommendation, alert).

	The teacher stresses that training data shapes outputs: unfair data gives unfair results (CS.MS.5.2). **Blueprint toolkit – flowchart symbols** (3 mins) Start/End, Input/Output, Process, Decision (Yes/No) and the class "safety/fairness check" symbol. Arrows show the direction of flow. **Worked example** (5 mins) Draft blueprint of the Lost & Found Helper, including a privacy check and a decision with Yes/No paths. The teacher asks where a human-in-the-loop step could go. **Task 2 – Draft your AI blueprint** (20 mins) – Worksheet Parts C–E Students complete the IPO table (3+ inputs, 2+ outputs, training data source). They then draw a flowchart (Option A) or a 6-panel storyboard, "A Day in the Life of Your AI" (Option B), with 5+ steps, one decision and one safety/fairness check. Teacher check-in questions: Where does your data come from? Could the AI treat any student unfairly? What happens if the AI is unsure? The teacher initials each draft once the MUST criterion is met. **Peer feedback – Two stars and a wish** (5 mins) – Worksheet Part F Partners swap worksheets, tick the 5-point blueprint checklist and write two stars and one wish.
Time	45 min
Plenary	**Exit Ticket 3-2-1 + confidence check** • Name 3 data inputs your AI uses • Name 2 outputs your AI produces • Describe 1 safety or fairness check and who it protects Students rate themselves 1–3 against MUST/SHOULD/COULD.
Time	5 min
Resources	Lesson 2 PowerPoint (W9.2), Lesson 2 student worksheet, Lesson 1 completed worksheets, Fix-the-Flow step cards, cut-out step cards (Support), rulers, coloured pencils, devices, Toddle.

Differentiation	

Support	• **L1:** choose a problem from the GRASPS example cards; use picture cards to pick two AI types; sentence starters ("Idea A uses ___ to ___") and • **L2:** cut-out step cards to order the AI's steps; 4-box storyboard frame with sentence starters; drawing allowed instead of writing.
Challenge	• **L1:** explain a trade-off (e.g. accuracy vs privacy) and predict how each idea could produce a biased result, with a fix. • **L2:** add a fairness/safety check and explain how it protects users; add a "human-in-the-loop" path for when the AI is unsure.
IEP	As per individual IEP accommodations (extra time, reduced written output, seating near teacher, chunked instructions).
ILP	• Worksheets are differentiated with Support / Core / Challenge levels, sentence starters, word banks and visual cues. • Students may draw a simple diagram or storyboard instead of writing full sentences. • Key vocabulary (design idea, feasibility, trade-off, blueprint, input/output) is pre-taught with visuals.
Gifted & Talented	• **L1:** mini case study – how would each idea cope with an unexpected situation (e.g. a new student in week one)? • **L2:** list the hardware, data and software requirements needed to build the chosen AI (Criterion B strand iv).

Homework

- **Lesson 1:** Upload photos of the two annotated ideas, comparison table and feasibility scorecard to Toddle.
- **Lesson 2:** Act on your peer's "wish", then upload your draft blueprint/storyboard to Toddle.

Reflection Notes

Lesson 1 (W9.1):

Lesson 2 (W9.2):